Razeen Wasif

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Summary

Software Engineer & ML Researcher with a strong foundation in **AI, machine learning,** and **software construction**. My background in physics provides a rigorous analytical and quantitative approach to problem-solving. Experienced in designing and implementing AI models, building **training infrastructure**, and shipping **full-stack platforms** using **Python, C++, CUDA,** and **TypeScript**.

Professional Experience

ANU Virtuous Loop
Full-Stack Developer & UX Lead — Feb 2026 – Present

- Collaborating with an industry client to architect a scalable feedback dashboard on the **Microsoft Power Platform (Power Apps, Power BI, Dataverse)**.
- Spearheading a complete **UI/UX overhaul** (user research + wireframing), lifting navigation efficiency and data accessibility by 40%.
- Engineering automated **Power Automate** pipelines to replace legacy manual processes with real-time feedback sync.
- Implementing **Row-Level Security (RLS)** and enterprise data validation to meet ANU's privacy and compliance standards.

Projects

Prism: GPU-Accelerated AutoML & Entity Resolution Engine
Live / Github

- Engineered a **GPU-first AutoML pipeline** for heterogeneous tabular data — automated EDA, target-aware cleaning, and **Optuna**-powered hyperparameter optimisation.
- Integrated a **RAPIDS**-accelerated record linkage workflow for entity resolution, using async execution and semantic embeddings to maximise candidate pair recall.
- Developed custom CUDA-backed XGBoost and PyTorch wrappers with automated CUDA/MPS/CPU fallback, served via a **FastAPI + Docker + React/Vite** stack and an Ollama LLM for NL data exploration.

Flux: AI-Native Desktop Browser — Github

- Built an **AI-native desktop browser** in **Rust + Tauri v2** (native OS webviews, no Electron) — an **8–15 MB** binary with <80 MB idle RAM, using a **zero-copy DOM IPC pipeline** that shares page snapshots as `Arc<[u8]>` across the terminal, agent, and embedder.
- Integrated a fully **local LLM agent** (Gemma 12B via Ollama / llama.cpp) that reads the page DOM and performs preview-and-approve actions behind a destructive-action deny-list — no cloud, no token cost.
- Shipped a real-**PTY** terminal sidebar with a DOM-aware CLI, webview hibernation under memory pressure, a Brave-engine ad/tracker blocker, and an **AES-256-GCM** password vault with OS-keychain key storage.

Omni: From-Scratch Hybrid Search Engine — Github

- Hand-rolled a search engine in **std-only Rust** (Go crawler, Python eval harness) — inverted index, **BM25F** ranking, Block-Max WAND pruning, Porter stemming, and a hand-written **HNSW** ANN index, with no Lucene / Tantivy.
- Fused lexical and **768-dim semantic** retrieval via Reciprocal Rank Fusion with passage-level scoring, reaching **0.666 nDCG@10** (vs. 0.477

Education

Australian National University
Bachelor of Advanced Computing (Honors)

- **Coursework:** Machine Learning, Artificial Intelligence, Computer Vision, Software Engineering, Computer Architecture, Algorithms and Data structures, Data analysis

Open Source Contributions

Pygame Library (GitHub)

- Identified and resolved a **segmentation fault** in the `PixelArray` module — debugged memory management in the **C backend** to prevent invalid access during array slicing.

Research

Quantized Domain-Specialist Verifiers for Multi-Agent LLM Scientific Reasoning — Github

- Investigating a hybrid architecture where frontier API models drive multi-agent debate (**Council** '/discover') and locally-hosted, fine-tuned, aggressively-quantized open-weights models act as a **domain-specialist verification layer** for domain-specific claims.
- Built a **single-GPU LoRA + DoRA + QLoRA pipeline** (Unsloth, NEFTune, sequence packing, 8-bit paged AdamW) on Mistral-7B-Instruct-v0.3 — producing physics / math / CS specialists trained partly on **Council-generated synthetic debate data**.
- Designing a controlled ablation across six bit-depths (FP16 / Q8 / Q6 / Q4 / Q3 / Q2) comparing **PTQ vs. QAT** in the specialist-as-verifier regime; target venue: NeurIPS ENLSP / ICLR ME-FoMo / ICML AI4Science 2026.

Partial Fine-Tuning of BioCLIP 2.5 for Multi-Label Plant ID (PlantCLEF 2026) — Project / Kaggle

lexical-only) and **0.97 success@10** on a 12k-document graded benchmark.
- Engineered live incremental indexing with content-addressed segments, a background tiered-merge thread, and atomic `Arc` snapshot swaps (zero downtime); serves a custom **HTTP/1.0 + SSE** API with local-LLM RAG answers that powers the Flux omnibox.

Nexus: Distributed Training Console Live

- Architected a standalone control plane and real-time monitoring console for distributed ML training, with its own backend, brand, and deployment surface.
- Built a **Firestore-backed** pod inventory and run-lifecycle model, surfacing NCCL liveness probes and per-rank telemetry rolled up live in a monochrome glass shell.
- Designed a streaming event console for multi-rank orchestration — per-step throughput, gradient norms, and rank-level failures without polling — on **React + TypeScript + Firebase** with a Python ingestion service.

Oracle: Multi-Domain Identification Platform Live

- Engineered a live multi-domain identification platform that productizes the **BioCLIP 2.5** PlantCLEF model — image upload, top-k predictions with taxonomy context, served via a **React + TypeScript + Firebase** front end.
- Implemented the inference path with **4×4 tile decomposition** and multi-resolution (224 / 336 px) ensembling, with logit-adjusted adaptive thresholding tuned for long-tail multi-label outputs.
- Validated end-to-end on the **PlantCLEF 2026** benchmark (7,806-class long-tail), reaching **0.418 Macro F1** on the held-out vegetation-plot test set.

Terminal Workbench: Onyx · AudioPulse · BoxTube Onyx / AudioPulse / BoxTube

- Designed and shipped three day-in-the-life TUIs in a four-day sprint — markdown notes, music, and video — each rendering its native media inside the terminal.
- **Onyx** (**Rust** / Ratatui): Obsidian-style markdown vault with `[[wikilinks]]` + backlinks, command palette, full-vault content search, ASCII graph view, and vim-style modal editing.
- **AudioPulse** (**Go** / Bubble Tea): Spotify-style music player playing full songs through an embedded **librespot** device (PKCE OAuth via Spotify Web API), album art rendered as 24-bit half-blocks.
- **BoxTube** (**Python** / Textual): YouTube client with full video playback rendered inside the terminal via **mpv** and the **kitty graphics protocol** (sixel / unicode fallbacks), no Google Data API required.

Custom 16-Bit RISC CPU Design & Implementation github

- Designed and simulated a **16-bit RISC processor** from basic logic gates — custom ISA, multi-stage datapath (ALU, register file, microprogrammed control unit), and a 16-bit address bus with dedicated PC / IR.
- Validated end-to-end through unit tests of individual components and full assembly-level programs (Fibonacci, array traversal).

- Partial fine-tune of **BioCLIP 2.5 ViT-H/14** with auxiliary **genus / family / order / class** heads, injecting taxonomic structure into the loss to stabilise training on a 7,806-class long-tail benchmark.
- Engineered a **4×4 tile inference** pipeline ensembled across **224 and 336-pixel** resolutions, with **logit-adjusted adaptive thresholding** per class to handle multi-label vegetation plots.
- Achieved **0.418 Macro F1** on the PlantCLEF 2026 leaderboard; paper submitted to **LifeCLEF / CLEF 2026** (ANU at PlantCLEF 2026).

Signal-Preserving TinyML for Gravitational Wave Triggers github

- Investigated the impact of **INT8 quantization** on phase-coherence in Gravitational Wave (GW) signals for deployment on ultra-low-power **ARM Cortex-M** hardware.
- Developed a ”Signal-Preserving” framework using **Quantization-Aware Training (QAT)** and Hardware-Aware Distillation to minimize SNR loss in stochastic astrophysical data.
- Engineered custom **C++ inference kernels** bypassing TensorFlow Lite Micro for 4096 Hz strain data within a **¡128KB SRAM** budget, achieving over 99% data compression at the target FAR.

Technologies

Languages: Python, C++, CUDA, Java, TypeScript, ARM ASM, SQL

ML / AI: PyTorch, TensorFlow, Optuna, RAPIDS, BioCLIP, DINOv3

Systems: Linux Kernel, TensorRT, NVIDIA DALI, FastAPI, Docker

Platforms: Microsoft Power Platform, Power BI, Dataverse, Firebase, Figma, React

Certifications

Deloitte Australia Technology Job Simulation
Forage *Aug 2024*

- Data analysis for Deloitte’s Technology team — Tableau dashboard, client proposal, Excel-based classification and business conclusions.